ANALYTICAL CUMULATIVE EXAM

March 5, 1994

Describe in as much detail as you can the work reported in a paper published in one the following journals, and received by our library in the last 12 months:

Analytical Chemistry

Analytica Chimica Acta

Sensors and Actuators

Talanta

Geochimica Cosmochimica Acta

Journal of Chromatography

Microchemical Journal

Journal of Chromatographic Science

Journal of the Electrochemical Society

Langmuir

Journal of Electroanalytical Chemistry

Conditions of selection: The paper you select must deal with a topic in Analytical Chemistry. In addition, the paper chosen must <u>not</u> be one that was published by a member of your own research group.

Make sure you identify the paper as much as possible:

journal (required)

main authors

research group or affiliation

approximate month of appearance or page number of current issue

Analytical Cumulative Examination

Fluorescence spectroscopy is an extremely sensitive way to detect and study the interactions of specific molecules. This problem is designed to investigate the capabilities of the fluorescence technique.

1. Recent reports have suggested that properly designed fluorescence measurements are capable of detecting a single molecule (fluorophor). Even in non-optimized situations, ultra-sensitive detection has been demonstrated. Suppose that a sample was illuminated with 250 milliwatts of 488.0 nm light. The interaction length is 1 mm. The sample contains a fluorescent species with a molar absorptivity of 20,000 cm^{-1} M^{-1} and a quantum yield of 0.25. Light is collected with a 1 cm diameter, f/1 lens and directed into a PMT with the following characteristics: quantum yield, 0.10; gain, 1×10^7; fluctuations in the dark current (σ), 2 picoamp. Given the limitations imposed by the PMT and the fundamental parameters of the fluorophor, determine the minimum concentration detectable under these conditions.

2. Often this level of detection can not be achieved due to other processes which limit the fluorescence emission. One example of such a process is dynamic quenching in which contact (collision) between the excited lumophor and the quenching species, Q dissipates the energy through non-radiative processes. The fluorescence quantum efficiency in the absence of quenching, Φ_f^o, is given as

$$\Phi_f^o = \frac{k_f}{k_f + k_{nr}}$$

whereas, with quenching, the quantum yield, Φ_f, is given by

$$\Phi_f^o = \frac{k_f}{k_f + k_{nr} + k_q[Q]}$$

In the above equations, k_f is the rate constant for deactivation of the excited state through emission, and k_{nr} is the rate constant for deactivation through non-radiative processes. When quenching is present, k_{nr} consists of several terms, one of which is dependent upon the quencher concentration, that is

$$k_{nr} = k_{nr} + k_q[Q]$$

(a) From these relationships, derive the Stern-Volmer equation

$$\frac{\Phi_f^o}{\Phi_f} = 1 + K_q[Q]$$

where K_q is the Stern-Volmer quenching constant defined as $K_q = k_q/k_f + k_{nr}$ (in L mol^{-1})

(b) Rearrange your result from (a) to the form given below (show work)

$$(\Phi_f)^{-1} = (\Phi_f^o)^{-1} + \frac{K_q[Q]}{\Phi_f^o}$$

(c) Using the data given below, and your result from (b), determine the Stern-Volmer Quenching constant for this quencher (with correct units). Note that the observed fluorescence intensity, I_f is proportional to the fluorescence quantum yield.

Normalized I_f	$[Q] \times 10^4$ M
1.0	2
0.63	4
0.44	6
0.32	8
0.27	10